

Embedding and Embedding Finetuning

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3. Keyword vs Semantic Search(Similarity Search)
4. Encoding vs Embedding
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What is an Embedding?

Embedding is mathematical representation of the data

Embedding is a way to convert text, image, audio, video into meaningful numbers.

Embedding is dense, high-dimensional numerical vector that represents the semantic meaning of data.

Dense: Non Spars

High-dimensional: 384, 768, 1024, 1536, 2048 etc....

vector : Mathematical object in linear algebra In programming this vector is typically stored as an array, means a collection of numerical values(float).

The capital of France is Paris →	0.10	-34	-52	-77	-0.40	-0.28	-0.19
Berlin is Germany's capital →	0.00	-35	-52	-065	-0.91	-0.03	-0.29

What is Semantic Search?

Semantic search = Meaning-based intelligent search using embeddings.

Semantic search means searching based on meaning, not just exact words.

Semantic search is usually performed using:

- ✓ Cosine similarity
- ✓ Dot product
- ✓ Euclidean

Example:

- eating fiber reduces heart risk
- eating fruits and vegetables lowers cardiovascular disease chances

Even though the words are different, embeddings place them close together in vector space.

Both sentences have the same meaning.

That's why semantic search returns relevant results even when wording changes.

Example:

- Eating fiber reduces heart risk
- Buying a new car improves driving comfort

Completely different topics

Health vs Automobile

Embeddings will be far apart

Low cosine similarity

Keyword vs Semantic Search(Similarity Search)

Clean Comparison Line

Keyword search matches words.

Semantic search matches meaning: find out similarity using matrix like cosine, Euclidian, dot product etc.

How Keyword-Based Search Work

If you search:

"How to reduce heart risk"

The system will mainly look for documents containing those exact words like:

- reduce
- heart
- risk

If a document instead contains:

"Ways to lower cardiovascular disease chances"

A traditional keyword search may not match it because:

- reduce $\neq$ lower
- heart risk $\neq$ cardiovascular disease

Even though the meaning is similar, the words are different

Example

Doc1: how to lose weight fast

Doc2: quick fat loss strategies

Keyword search $\rightarrow$ weak match

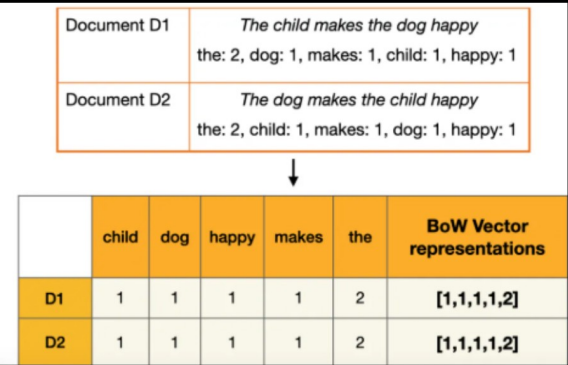
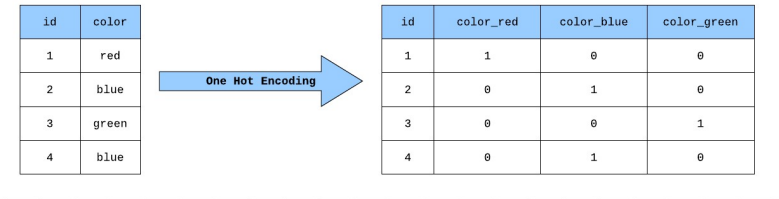
Semantic search $\rightarrow$ strong match

Encoding vs Embedding

Encoding: Converts data into a numerical format so that machines can process it.

Count based, matrix factorization based

One-hot encoding, BOW, TF-IDF, BM25, GloVe



$$TF(t,d) = \frac{\text{number of times } t \text{ appears in } d}{\text{total number of terms in } d}$$
$$IDF(t) = \log \frac{N}{1 + df}$$
$$TF-IDF(t,d) = TF(t,d) * IDF(t)$$

Embedding: Converts data into a numerical vector that preserves meaning .

Embedding("king") Embedding("queen") Distance between them → small Angle will be less between two vector

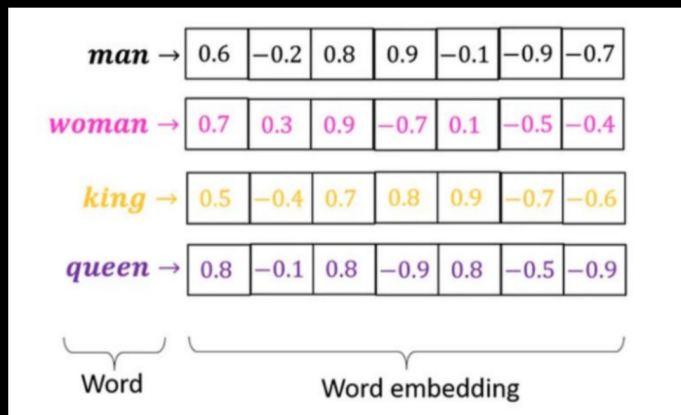
Embedding("king") Embedding("banana") angle between huge Distance → large

Classical Word Embedding Models

- Word2Vec, , FastText

Limitation:

- They create embeddings at the word level only.
- They produce static embeddings (fixed vectors).
- They do not understand context.



SOTA Transformer-based Embeddings Model

- Based on Transformer architecture
- Contextual embeddings Context-aware embeddings.
- Any text (word, phrase, or sentence, Full paragraphs)

Example: Sentence Transformers Models, MiniLM, BGE, OpenAI Embeddings, Gemini Embedding, Claude Embedding etc....

The capital of France is Paris →	0.10	-34	-52	-77	-0.40	-0.28	-0.19
Berlin is Germany's capital →	0.00	-35	-52	-0.65	-0.91	-0.03	-0.29

Example

Sentence 1: I sat on the river bank.

Sentence 2: I deposited money in the bank.

Word: bank

In a static embedding model(word2vec), the word "bank" is represented by the same vector in both sentences.

bank → [0.21, -0.44, 0.89, ...]

In SOTA Transformer-based Embeddings Model , the word "bank" will be represented by the different vector.

bank → [0.21, -0.44, 0.89, ...]

bank → [0.93, 0.01, 0.75, ...]

Embedding Models

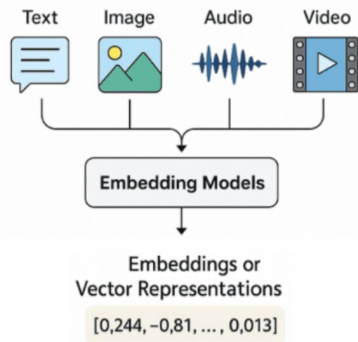
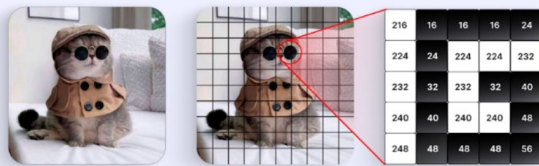
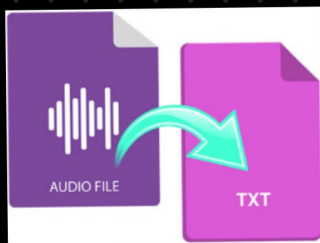


Image Embeddings: We can also convert images into embeddings using Vision or Multimodal models (CLIP, Vision Transformer(ViT))



Approach 1: Audio → Text → Text Embedding



Audio → ASR (Speech-to-Text) → Text → Text Embedding Model → Vector DB

OpenAI Whisper

Google Speech-to-Text

Meta MMS

Approach 2: Direct Audio Embedding (Advanced)

Audio → Audio Encoder Model → Audio Embedding → Vector DB



Google YAMNet

OpenAI CLAP

Meta wav2vec

Audio Embeddings

Use Cases

- Voice search
- Music similarity

Video Embeddings

Video = sequence of images + audio + time dimension

Video

- └─> Audio → ASR → Text → Text Embedding
- └─> Frames → Vision Encoder → Image Embedding
- └─> Metadata → Text Embedding

→ Multi-vector Retrieval

Direct Video Encoder

Video → Video Transformer Encoder → Video Embedding → Vector DB

Video Embedding Models

VideoMAE

TimeSformer

I3D

Use Cases

Video search

Content moderation

Recommendation (YouTube / Netflix)

Application of embedding

1. Semantic Search (Most Fundamental): Meaning-based search
Instead of matching keywords, we match vector similarity.
2. Google-like search engines
3. Google Image search
4. Pinterest

User Search: "a red car" System retrieves: "Images of red cars"

5. Recommendation Systems

Netflix: Similar movies

Amazon: Similar products

Spotify: Similar songs

All use vector similarity.

6. Topic Modeling (Clustering)
7. Group similar documents together

News categorization

8. RAG (Retrieval-Augmented Generation)

Models for Embedding

Model Name	Modality	Provider	Category
text-embedding-3-small	Text	OpenAI	Closed
text-embedding-3-large	Text	OpenAI	Closed
text-embedding-ada-002 (legacy)	Text	OpenAI	Closed
Gemini Embeddings	Text	Google	Closed
Claude Embeddings	Text	Anthropic	Closed
Cohere embeddings	Text	Cohere	Closed
Titan Text Embeddings	Text	AWS (Amazon Bedrock)	Closed

all-mpnet-base-v2	Text	HuggingFace (Sentence-Transformers)	Open
all-distilroberta-v1	Text	HuggingFace (Sentence-Transformers)	Open
all-roberta-large-v1	Text	HuggingFace (Sentence-Transformers)	Open
msmarco-minilm-v3	Text	HuggingFace (Sentence-Transformers)	Open
BAAI-BGE-Base-En	Text	HuggingFace (BAAI)	Open
BAAI-BGE-Large	Text	HuggingFace (BAAI)	Open
BGE-multilingual	Text	HuggingFace (BAAI)	Open
e5-small	Text	HuggingFace	Open
e5-base	Text	HuggingFace	Open
e5-large	Text	HuggingFace	Open
st5-xd-embed	Text	HuggingFace	Open
miniLM L6	Text	Microsoft(OpenAI)	Open
miniLM L12	Text	Microsoft(OpenAI)	Open
OpenCLIP	Image (Vision)	HuggingFace (OpenAI)	Open
BLIP2 embedding models	Image (Vision)	Various	Open
DINO-v2 embeddings	Image (Vision)	Meta	Open
Wav2Vec 2.0	Audio	Meta	Open
HuBERT	Audio	Meta	Open
YAMNet	Audio	Google	Open
CLAP (Contrastive Language-Audio Pretraining)	Audio	LAION	Open
VideoMAE	Video	Meta	Open
I3D (Inflated 3D ConvNet)	Video	DeepMind	Open

How to select best embedding model

1. Embedding Quality (Most Important): Check benchmark performance:

Look at:

- MTEB leaderboard
- BEIR benchmark

<https://arxiv.org/abs/2104.08663>

2. Dimensionality

Example:

- 384 dim → lightweight
- 768 dim → balanced
- 1536 dim → high quality but heavy

Higher dimension:

- Better semantic representation
- But more storage cost
- More memory usage
- Speed impact

3. Cost

Close source (Open AI embedding):

- Easy, high quality, API cost

Open source:

- Free but infra + scaling cost

4. Domain Suitability

General model:

- all-MiniLM
- OpenAI embedding

Embedding model leader board

MTEB Leaderboard (Hugging Face Space)

<https://huggingface.co/spaces/mteb/leaderboard>

This shows various embedding models ranked across different tasks.

MTEB Official Framework & Leaderboard Repo (GitHub)

<https://github.com/embeddings-benchmark/results>

MTEB Homepage & Info

<https://huggingface.co/mteb>

Explanation of the benchmark and how it evaluates models.

HF Sentence Transformer Doc: <https://www.sbert.net/>

ST Repo: <https://huggingface.co/models?library=sentence-transformers&sort=downloads>

Core Architectural Concept of an Embedding Model:

Text



Tokenizer



Transformer Encoder



Hidden States (per token vectors)



Pooling (convert token vectors → single sentence vector)



(Optional Dense layer / projection)



Final Embedding Vector(e.g., 384 / 768 / 1024 / 1536 dims)

Embedding Model vs GPT Model

Embedding Model

Generates vector representation

Encoder-focused (mostly)

Pretrain on:

Large web crawl

Web-scale text corpus

Embedding models are trained(supervisory tuned) explicitly to structure vector space such that:

- Similar text → close vectors
- Dissimilar text → far vectors

Embedding models normally train hote hain:

Contrastive Learning

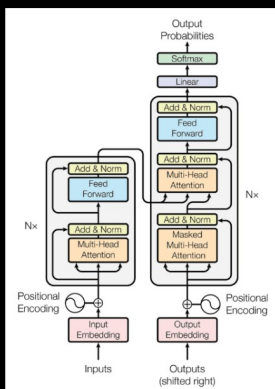
Goal:

Similar sentences → close vectors

Different sentences → far vectors

Example loss:

- Triplet loss
- Cosine similarity loss
- InfoNCE
- Multiple negatives ranking loss



Architecture Category	Backbone Type	Retrieval Setup	Used For	Example
Encoder-Only Model	BERT-style	Used in Dual Encoder retrieval	Most embedding models	SBERT, E5, BGE
Dual Encoder Setup	Encoder backbone	Separate encoding of query & doc	Scalable RAG	SBERT retrieval
Cross Encoder	Encoder backbone	Joint encoding of <u>query+doc</u>	Re-ranking	M5-MARCO cross models
Decoder-Based Embedding	GPT-style	Pool hidden states	Commercial APIs	OpenAI, Gemini (likely)
Multimodal Dual Encoder	Separate encoders per modality	Contrastive alignment	Image-text search	CLIP

How Does It Work?

Typical embedding model training uses:

Contrastive Learning

Pairs of similar/dissimilar text are used

Goal:

similar → close vectors

different → distant vectors

This is usually done with:

cosine similarity

triplet loss

contrastive loss (InfoNCE)

During fine-tuning, the model adjusts internal weights to produce embeddings that are more aligned with your own data semantics.

Pretraining Data

Usually mixture of:

Wikipedia
Common Crawl
BooksCorpus
Web text
News articles
Open datasets
Public web pages

Embedding-Specific Fine-Tuning

MS MARCO (Query-Passage pairs)
Natural Questions
NLI datasets (SNLI, MNLI)
STS Benchmark
BEIR benchmark datasets
Reddit Q&A pairs
QA pairs
Instruction-response pairs (E5 style)

<https://huggingface.co/datasets/sentence-transformers/all-nli>

<https://huggingface.co/datasets/sentence-transformers/stsb>

<https://huggingface.co/datasets/sentence-transformers/quora-duplicates>

<https://huggingface.co/datasets/sentence-transformers/natural-questions>

Massive Unlabeled Text → Pretraining

↓

Labeled Similarity / Retrieval Data → Contrastive Fine-Tuning

↓

Final Embedding Model

Data set format for the embedding finetuning?

Inbuilt HuggingFace Embedding Dataset

<https://huggingface.co/datasets/sentence-transformers/all-nli>

<https://huggingface.co/datasets/sentence-transformers/stsb>

<https://huggingface.co/datasets/sentence-transformers/quora-duplicates>

<https://huggingface.co/datasets/sentence-transformers/natural-questions>

Case A — Pair Format

```
{
  "sentence1": "text A",
  "sentence2": "text B",
  "label": 1    # 1 = similar, 0 = not similar
}
```

Case B — Triplet Format (Anchor / Pos / Neg)

```
{
  "query": "how to bake cake",
  "positive_passage": "...",
  "negative_passage": "...
}

{
  "anchor": "some text",
  "positive": "similar meaning text",
  "negative": "unrelated text"
}
```

Case C — N-way format (Ranking)

```
{
  "query": "question",
  "positives": ["doc1", "doc2"],
  "negatives": ["docA", "docB", "docC"]
}
```

What is embedding finetuning

Embedding fine-tuning means training an embedding model further on your specific data so that its vector representation becomes more accurate and relevant for your domain or task.

Embedding fine-tuning means training an embedding model so that its vector representations become more task-specific rather than generic.

Normal Embeddings

- Pretrained on large generic data (Wikipedia, Common Crawl, QA pairs)
- Good for general search/semantic similarity

Fine-Tuned Embeddings

- Further trained on your specific dataset
- Model learns better semantic structure for your domain

So instead of using the original "out-of-the-box vectors," you customize them for your use-case.

why we should do the embedding finetuning

To get a specialized embeddings model or domain specific or to get a task specific model

Here are the real reasons based on practical use-cases:

1. Domain-Specific Understanding

You have domain language like:

- Legal contracts
- Medical reports
- Finance docs
- Product descriptions

Generic embeddings treat all language alike.

Fine-tuned embeddings understand domain semantics better.

2. Better Search / Semantic Retrieval

If you are building:

- RAG (Retrieval Augmented Generation)
- Vector DB search
- Document QA

Fine-tuned embeddings give higher recall + precision because they are optimized for your corpus.

3. Task-Specific Nearness

Examples:

- Customer support pairs → label vs query
- FAQ matching
- Intent detection
- Duplicate detection

Fine-tuning aligns vectors with task intent not just basic language semantics.

Aspect	Embedding Fine-Tuning	LLM Fine-Tuning
Core Purpose	Improve semantic vector space	Customize text generation
Training Focus	Similarity & ranking	Next token generation
Loss Function	Contrastive / Triplet / Ranking	Cross-Entropy
Dataset	Paired / Triplet	Chat / Instruction
Output	High-quality embeddings	High-quality responses
In RAG	Improves retrieval	Improves generation
Architecture	Encoder-only	Decoder-heavy / Causal

